

FIREWALL FUNDAMENTALS:

- A firewall is a guard which checks the network's or digital device incoming and outgoing traffic and blocks the traffic which found unauthorized or suspicious on the basis of rules you have maintained in it.

- There are many types of firewalls, each has its unique purpose and works on different layers of OSI model. Some of the most common firewalls are:

i) Stateless firewall: It works on layer 3 and layer 4 of the OSI. In this type of firewall, the firewall filters data based on the rules and it doesn't check the state of previous connections. Ex: Suppose a firewall blocked few packet from a single source, ideally it should drop all the future packets from this source but it doesn't save the previous connection data so its future packets from this source will be treated as new and checked again on basis of rule. Efficient for high speed networks.

ii) Stateful firewall: It works on layer 3 and layer 4 of the OSI model. In this type of firewall, it keeps a record of previous connections made and stores it in state table. It adds a layer of security based on their history with connections. If the previous packets from the connection are denied the new one from the same will be denied and if the previous connection was stable, new packets will be allowed.

iii) Proxy firewall: It works on the layer 7 of the OSI model. In this type of firewall we don't directly talk to a website.

- Stateful and Stateless firewall doesn't inspect the content of the packet it only checks source IP, ports etc but Next Generation and Proxy inspects the content of the packet also.

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or anything, firewall acts as a middleman who talks to the website for you and ~~you~~ provides ~~anonymity~~ anonymity for the internal IP addresses. These firewalls also inspect the content of the packets and filtering rules can be applied to them based on ~~their contents~~ which they allow/deny traffic on basis of their content. It also decrypts and inspects SSL/TLS data packets.

iv) Next Generation firewalls: This is the most advanced type of firewall. It works on layer 3 to 7 of the OSI model. It offers deep packet analysis functionality and also has intrusion prevention system which blocks malicious activities in real time. It also has SSL/TLS decryption capabilities, which decrypts the packets and correlate the data with threat intelligence feeds to make efficient decisions.

• Rules in firewalls:

A firewall filters traffic based on its built-in rules, but some customized rules can also be defined for our network. The basic component for firewall's rules are described below:

- a) Source address: From where the traffic originates
- b) Destination address: ~~From where~~ The machine that would receive traffic
- c) Port: The port no. for the traffic
- d) Protocol: Protocol used during the communication
- e) Action: Defines the action that would be taken, for traffic of this nature
- f) Direction: It defines rule's applicability to incoming or outgoing traffic.

• Types of actions:

- a) Allow: A rule's "allow" action indicates that all the traffic defined inside the rule would be permitted.

b) Deny: It deny all the traffic that falls under the rule's defined.

c) Forward: It forwards or we can say redirect the traffic to a different network segment.

• Directionality of rules:

i) Inbound rules: Rules are said to be inbound rules when they are applied only to incoming traffic.

ii) Outbound rules: Rules are said to be outbound rules when they are applied only to outgoing traffic.

iii) Forward rules: Forwarding rules are created to forward the specific traffic inside the network.

• Windows Defender Firewall: It is a built-in firewall introduced by Microsoft in the windows OS. In this firewall we can allow, deny specific programs and create customized rules.

The windows defender firewall homepage shows the "Network profile" and the available options.

• a) Network profiles: There are two available network profiles.

i) Private network: It contains the firewall configurations which work when connected to a private network such as home network.

ii) Guest or Public network: It contains the firewall configurations that work when connected to a Guest or untrusted network like coffee shops, restaurants etc.

• iptables, ~~iptables~~ nftables, firewalld etc, are the tools to control the netfilter firewall framework.

iptables/nftables/firewalld → gives rules
netfilter → applies rules on packets.

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- Custom rules can be created using the advanced setting option in the home dashboard. Then we have to select whether we want to create a outbound or inbound rule and then select new. Then select custom rules.

- **Linux iptables firewall:**

Linux OS also offers the functionality of a built-in-firewall. We have many firewalls here. Let's briefly review most of them:

i) **Netfilter:** It is a framework inside the Linux OS ~~os~~ that provides firewall functionalities including packet filtering, NAT, and connection tracking. Some common firewall utilities that use this framework are:

a) **iptables:** It uses the netfilter framework that provides various functionalities to control network traffic.

b) **nftables:** It is a successor to the iptables with enhanced packet filtering and NAT capabilities.

c) **firewalld:** This utility has predefined rule sets. It comes with different pre-built network zone configurations.

- **ufw:** It stands for uncomplicated firewall. It is another tool or utility which works on netfilter framework. It helps you to make rules using simple commands unlike iptables which requires complex commands to make rules. You can make rules using simple commands via ufw which will then be configured into your desired rules in iptables.

commands making rules using

- Some of the ~~rules~~ for `ufw` are:

a) `sudo ufw default allow outgoing`: It allows all the outgoing ~~com~~ traffic from our machine.

Here `allow` can be changed to `deny` and outgoing can be changed to incoming based on the requirement of the rule we want to create.

b) `sudo ufw deny 22/TCP`: Deny incoming traffic at any port on your system. In this we block the incoming SSH connection.

c) `sudo ufw status numbered`: Lists all the active rules in numbered order.

d) `sudo ufw delete number`: Delete any rules by placing its rule number in place of number in the command.

NOTE: Default in the first command means that it works by default for all traffic unless it is mentioned ~~to deny~~ in another to deny outgoing traffic in some other rules.